Size : 360x230mm

45gsm with fold

CYTOMIB

Bortezomib 3.5 mg
Powder for Solution for Injection

PRODUCT DESCRIPTION
Description of finished product (Before dilution): *Sterile white to off-white lyophilized mass.*
Description of finished product (After dilution with 0.9% Sodium Chloride solution): *The colour of solution is clear.*

PHARMACODYNAMICS
Pharmacotherapeutic group: Antineoplastic agents, other antineoplastic agents
ATC code: L01XX32.

Mechanism of action
Bortezomib is a proteasome inhibitor. It is specifically designed to inhibit the chymotrypsin-like activity of the 26S proteasome in mammalian cells. The 26S proteasome is a large protein complex that degrades ubiquitinated proteins. The ubiquitin-proteasome pathway plays an essential role in regulating the turnover of specific proteins, thereby maintaining homeostasis within cells. Inhibition of the 26S proteasome prevents this targeted proteolysis and affects multiple signalling cascades within the cell, ultimately resulting in cancer cell death.
Bortezomib is highly selective for the proteasome. At 10 μM concentrations, bortezomib does not inhibit any of a wide variety of receptors and proteases screened and is more than 1,500-fold more selective for the proteasome than for its next preferable enzyme. The kinetics of proteasome inhibition were evaluated in vitro, and bortezomib was shown to dissociate from the proteasome with a t½ of 20 minutes, thus demonstrating that proteasome inhibition by bortezomib is reversible.
Bortezomib mediated proteasome inhibition affects cancer cells in a number of ways, including, but not limited to, altering regulatory proteins, which control cell cycle progression and nuclear factor kappa B (NF-kB) activation. Inhibition of the proteasome results in cell cycle arrest and apoptosis. NF-kB is a transcription factor whose activation is required for many aspects of tumourigenesis, including cell growth and survival, angiogenesis, cell-cell interactions, and metastasis. In myeloma, bortezomib affects the ability of myeloma cells to interact with the bone marrow microenvironment.
Bortezomib is cytotoxic to a variety of cancer cell types and that cancer cells are more sensitive to the pro-apoptotic effects of proteasome inhibition than normal cells. Bortezomib causes reduction of tumour growth in many tumour models, including multiple myeloma.

PHARMACOKINETICS
Absorption
Following IV bolus administration of a 1 mg/m^2 and 1.3 mg/m^2 dose to patients with multiple myeloma, the mean 1st-dose maximum plasma concentrations of bortezomib were 57 ng/mL and 112 ng/mL, respectively. In subsequent doses, mean maximum observed plasma concentrations ranged from 67-106 ng/mL for the 1 mg/m^2 dose and 89-120 ng/mL for the 1.3 mg/m^2 dose. The mean elimination half-life of bortezomib upon multiple dosing ranged from 40-193 hrs. Bortezomib is eliminated more rapidly following the 1st dose compared to subsequent doses. Mean total body clearances were 102 L/hr and 112 L/hr following the 1st dose for doses of 1 mg/m^2 and 1.3 mg/m^2, respectively, and ranged from 15-32 L/hr following subsequent doses for doses of 1 mg/m^2 and 1.3 mg/m^2, respectively.
Distribution
The mean distribution volume of bortezomib ranged from 1659-3294 L (489-1884 L/m^2) single- or repeat-dose administration of 1 or 1.3 mg/m^2 to patients with multiple myeloma. This suggests that bortezomib distributes widely to peripheral tissues. The binding of bortezomib to human plasma proteins averaged 83% over the concentration range of 100-1000 ng/mL.
Biotransformation
Human liver microsomes and human cDNA-expressed cytochrome P-450 isozymes indicate that bortezomib is primarily oxidatively metabolized via cytochrome P-450 enzymes, 3A4, 2C19 and 1A2. Bortezomib metabolism by CYP2D6 and 2C9 enzymes is minor. The major metabolic pathway is deboronation to form 2 deboronated metabolites that subsequently undergo hydroxylation to several metabolites. Deboronated-bortezomib metabolites are inactive as 26S proteasome inhibitors.
Elimination
The pathways of elimination of bortezomib have not been characterized in humans.
Special populations
Age, Gender and Race: The effects of age, gender and race on the pharmacokinetics of bortezomib have not been evaluated.
Hepatic impairment
The effect of hepatic impairment on the pharmacokinetics of bortezomib was assessed in cancer patients at bortezomib doses ranging from 0.5-1.3 mg/m^2. When compared to patients with normal hepatic function, mild hepatic impairment did not alter dose-normalised bortezomib AUC. However, the dose normalized mean AUC values were increased by approximately 60% in patients with moderate or severe hepatic impairment. A lower starting dose is recommended in patients with moderate or severe hepatic impairment, and those patients should be monitored closely.
Renal impairment
A pharmacokinetic study was conducted in patients with various degrees of renal impairment who were classified according to their creatinine clearance values (CrCl) into the following groups:
Normal (CrCl ≥60 mL/min/1.73 m^2, n=12), mild (CrCl 40-59 mL/min/1.73 m^2, n=10), Moderate (CrCl 20-39 mL/min/1.73 m^2, n=9) and; Severe (CrCl <20 mL/min/1.73 m^2, n=3).
A group of dialysis patients who were dosed after dialysis was also included in the study. Patients

were administered IV doses of bortezomib 0.7-1.3 mg/m^2 twice weekly. Exposure of bortezomib (dose-normalized AUC and Cmax) was comparable among all the groups.

INDICATION
CYTOMIB is indicated for the treatment of patients with multiple myeloma.
CYTOMIB is indicated for the treatment of patients with mantle cell lymphoma.

RECOMMENDED DOSE
Posology and Method of Administration
Important Dosing Guidelines
CYTOMIB is for **INTRAVENOUS OR SUBCUTANEOUS USE ONLY.** CYTOMIB should not be administered by any other route.
Because each route of administration has a different reconstituted concentration, caution should be used when calculating the volume to be administered.
The recommended starting dose of CYTOMIB is 1.3 mg/m^2. CYTOMIB may be administered intravenously at a concentration of 1 mg/ml, or subcutaneously at a concentration of 2.5 mg/ml.
CYTOMIB retreatment may be considered for patients with multiple myeloma who had previously responded to treatment with CYTOMIB and who have relapsed at least 6 months after completing prior CYTOMIB treatment. Treatment may be started at the last tolerated dose.
When administered intravenously, CYTOMIB is administered as a 3 to 5 second bolus intravenous injection.
Dosage in Previously Untreated Multiple Myeloma
CYTOMIB is administered in combination with oral mephalan and oral prednisone for nine 6-week treatment cycles as shown in Table 1. In Cycles 1-4, CYTOMIB is administered twice weekly (days 1, 4, 8, 11, 22, 25, 29 and 32). In Cycles 5-9, CYTOMIB is administered once weekly (days 1, 8, 22 and 29). At least 72 hours should elapse between consecutive doses of CYTOMIB.

Table 1. Dosage Regimen for Patients with Previously Untreated Multiple Myeloma

Twice weekly CYTOMIB (Cycles 1-4)												
Week	**1**				**2**		**3**	**4**		**5**		**6**
CYTOMIB (1.3 mg/m^2)	Day 1	--	--	Day 4	Day 8	Day 11	Rest period	Day 22	Day 25	Day 29	Day 32	Rest period
Melphalan (9 mg/m^2) Prednisone (60 mg/m^2)	Day 1	Day 2	Day 3	Day 4	--	--	Rest period	--	--	--	--	Rest period
Once Weekly CYTOMIB (Cycles 5 – 9 when used in combination with Melphalan and Prednisone)												
Week	**1**				**2**		**3**	**4**		**5**		**6**
CYTOMIB (1.3 mg/m^2)	Day 1	--	--		Day 8		Rest period	Day 22		Day 29		Rest period
Melphalan (9 mg/m^2) Prednisone (60 mg/m^2)	Day 1	Day 2	Day 3	Day 4	--	--	Rest period	--	--	--	--	Rest period

Dose Modification Guidelines for Combination Therapy with CYTOMIB, Melphalan and Prednisone.
Prior to initiating any cycle of therapy with CYTOMIB in combination with melphalan and prednisone:
• Platelet count should be at least 70 x 10^9/L and the absolute neutrophil count (ANC) should be at least 1 x 10^9/L.
• Non-hematological toxicities should have resolved to Grade 1 or baseline.

Table 2. Dose Modifications during Cycles of Combination of CYTOMIB, Melphalan and Prednisone Therapy.

Toxicity	Dose modifications
Hematological toxicity during a cycle: If prolonged Grade 4 neutropenia or thrombocytopenia or thrombocytopenia with bleeding is observed in the previous cycle.	Consider reduction of the melphalan dose by 25% in the next cycle.
If platelet count is not above 30 x 10^9/L or ANC is not above 0.75 x 10^9/L on a CYTOMIB dosing day (other than day 1).	Withhold CYTOMIB dose.
If several CYTOMIB doses in consecutive cycles are withheld due to toxicity.	Reduce CYTOMIB dose by 1 dose level (from 1.3 mg/m^2 to 1 mg/m^2 or from 1 mg/m^2 to 0.7 mg/m^2).
Grade 3 or higher non-hematological toxicities.	Withhold CYTOMIB therapy until symptoms of the toxicity have resolved to Grade 1 or baseline. Then, CYTOMIB may be reinitiated with 1 dose level reduction (from 1.3 mg/m^2 to 1 mg/m^2, or from 1 mg/m^2 to 0.7 mg/m^2). For CYTOMIB-related neuropathic pain and/or peripheral neuropathy, hold or modify CYTOMIB as outlined in Table 6.

For information concerning melphalan and prednisone, see manufacturer's prescribing information.
Dose modifications guidelines for peripheral neuropathy are provided.
Posology for previously untreated multiple myeloma patients eligible for haematopoietic stem cell transplantation (induction therapy).
Combination therapy with dexamethasone
CYTOMIB 3.5 mg powder for solution for injection is administered via intravenous or subcutaneous injection at the recommended dose of 1.3 mg/m^2 body surface area twice weekly for two weeks on days 1, 4, 8, and 11 in a 21 day treatment cycle. This 3 week period is considered a treatment cycle. At least 72 hours should elapse between consecutive doses of CYTOMIB.

Dexamethasone is administered orally at 40 mg on days 1, 2, 3, 4, 8, 9, 10 and 11 of the CYTOMIB treatment cycle.
Four treatment cycles of this combination therapy are administered.
Combination therapy with dexamethasone and thalidomide
CYTOMIB 3.5 mg powder for solution for injection is administered via intravenous or subcutaneous injection at the recommended dose of 1.3 mg/m^2 body surface area twice weekly for two weeks on days 1, 4, 8, and 11 in a 28 day treatment cycle. This 4 week period is considered a treatment cycle. At least 72 hours should elapse between consecutive doses of CYTOMIB.
Dexamethasone is administered orally at 40 mg on days 1, 2, 3, 4, 8, 9, 10 and 11 of the CYTOMIB treatment cycle.
Thalidomide is administered orally at 50 mg daily on days 1-14 and if tolerated the dose is increased to 100 mg on days 15-28, and thereafter may be further increased to 200 mg daily from cycle 2 (see Table 3).
Four treatment cycles of this combination are administered. It is recommended that patients with at least partial response receive 2 additional cycles.

Table 3: Posology for CYTOMIB combination therapy for patients with previously untreated multiple myeloma eligible for haematopoietic stem cell transplantation.

Cy+ Dx	**Cycles 1 to 4**				
	Week	**1**	**2**	**3**	
	Cy (1.3 mg/m^2)	Day 1, 4	Day 8, 11	Rest Period	
	Dx 40 mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-	
Cy+Dx+T	**Cycle 1**				
	Week	**1**	**2**	**3**	**4**
	Cy (1.3 mg/m^2)	Day 1, 4	Day 8, 11	Rest Period	Rest Period
	T 50 mg	Daily	Daily	-	-
	T 100 mg[a]	-	-	Daily	Daily
	Dx 40 mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-	-
	Cycles 2 to 4[b]				
	Cy (1.3 mg/m^2)	Day 1, 4	Day 8, 11	Rest Period	Rest Period
	T 200 mg[a]	Daily	Daily	Daily	Daily
	Dx 40 mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-	-

Cy=CYTOMIB ;Dx=dexamethasone; T=thalidomide
[a] Thalidomide dose is increased to 100 mg from week 3 of Cycle 1 only if 50 mg is tolerated and to 200 mg from cycle 2 onwards if 100 mg is tolerated.
[b] Up to 6 cycles may be given to patients who achieve at least a partial response after 4 cycles

Dosage adjustments for transplant eligible patients.
For CYTOMIB dosage adjustments, as described under **"Dosage and Dose Modifications for Relapsed Multiple Myeloma and Relapsed Mantle Cell Lymphoma"** and **"Dose Modifications for Peripheral Neuropathy"** should be followed.
In addition, when CYTOMIB is given in combination with other chemotherapeutic medicinal products, appropriate dose reductions for these products should be considered in the event of toxicities according to the recommendations in the local package inserts.

Dosage in Previously Untreated Mantle Cell Lymphoma.
CYTOMIB (1.3 mg/m^2) is administered intravenously in combination with intravenous rituximab, cyclophosphamide, doxorubicin and oral prednisone (VcR-CAP) for 3-week treatment cycles as shown in Table 4. CYTOMIB is administered first followed by rituximab. CYTOMIB is administered twice weekly for two weeks (Days 1, 4, 8 and 11) followed by a 10 day rest period on days 12-21.
For patients with a response first documented at cycle 6, two additional VcR-CAP cycles are recommended. At least 72 hours should elapse between consecutive doses of CYTOMIB.

Table 4. Dosage Regimen for Patients with Previously Untreated Mantle Cell Lymphoma

	Twice Weekly CYTOMIB (3-Week Cycles)*							
Week	**1**					**2**		**3**
CYTOMIB (1.3 mg/m^2)	Day 1	--	--	Day 4	--	Day 8	Day 11	Rest Period
Rituximab (375 mg/m^2) Cyclophosphamide (750 mg/m^2) Doxorubicin (50 mg/m^2)	Day 1	--	--			--	--	Rest Period
Prednisone (100 mg/m^2)	Day 1	Day 2	Day 3	Day 4	Day 5	--	--	Rest Period

*Dosing may continue for 2 more cycles (for a total of 8 cycles) if response is first seen at cycle 6

Dose Modification Guidelines for CYTOMIB When Given in Combination with Rituximab, Cyclophosphamide, Doxorubicin and Prednisone.
Prior to the first day of each cycle (other than Cycle 1):
• Platelet count should be at least 100 x 10^9/L and absolute neutrophil count (ANC) should be at least 1.5 x 10^9/L
• Haemoglobin should be at least 8 g/dL (at least 4.96 mmol/L)
• Non-hematologic toxicity should have recovered to Grade 1 or baseline

Interrupt CYTOMIB treatment at the onset of any Grade 3 hematologic or non-hematological toxicities, excluding neuropathy [see **Table 6** and **Warnings and Precautions**]. For dose adjustments, see **Table 5** below.

Table 5. Dose Modifications on Days 4, 8, and 11 during Cycles of Combination CYTOMIB, Rituximab, Cyclophosphamide, Doxorubicin and Prednisone Therapy.

Toxicity	Dose modifications
Hematological toxicity	
Grade 3 or higher neutropenia, or a platelet count not at or above 25 x 10^9/L.	Withhold CYTOMIB therapy for up to 2 weeks until the patient has an ANC at or above 0.75 x 10^9/L and a platelet count at or above 25 x 10^9/L.
	• If, after CYTOMIB has been withheld, the toxicity does not resolve, discontinue CYTOMIB. • If toxicity resolves such that the patient has an ANC at or above 0.75 x 10^9/L and a platelet count at or above 25 x 10^9/L, CYTOMIB dose should be reduced by 1 dose level reduction (from 1.3 mg/m^2 to 1 mg/m^2, or from 1 mg/m^2 to 0.7 mg/m^2).
Grade 3 or higher non-hematological toxicities.	Withhold CYTOMIB therapy until symptoms of the toxicity have resolved to Grade 2 or better. Then, CYTOMIB may be reinitiated with one dose level reduction (from 1.3 mg/m^2 to 1 mg/m^2, or from 1 mg/m^2 to 0.7 mg/m^2). For CYTOMIB-related neuropathic pain and/or peripheral neuropathy, hold or modify CYTOMIB as outlined in Table 6.

For information concerning rituximab, cyclophosphamide, doxorubicin and prednisone, see manufacturer's prescribing information.

Dosage and Dose Modifications for Relapsed Multiple Myeloma and Relapsed Mantle Cell Lymphoma.
CYTOMIB (1.3 mg/m^2/dose) is administered twice weekly for 2 weeks (Days 1, 4, 8, and 11) followed by a 10-day rest period (Days 12-21). For extended therapy of more than 8 cycles, CYTOMIB may be administered on the standard schedule or, for relapsed multiple myeloma, on a maintenance schedule of once weekly for 4 weeks (Days 1, 8, 15, and 22) followed by a 13-day rest period (Days 23 to 35). At least 72 hours should elapse between consecutive doses of CYTOMIB.
Patients with multiple myeloma who have previously responded to treatment with CYTOMIB (either alone or in combination) and who have relapsed at least 6 months after their prior CYTOMIB therapy may be started on CYTOMIB at the last tolerated dose. Retreated patients are administered CYTOMIB twice weekly (Days 1, 4, 8, and 11) every three weeks for a maximum of 8 cycles. At least 72 hours should elapse between consecutive doses of CYTOMIB. CYTOMIB may be administered either as a single agent or in combination with dexamethasone.
CYTOMIB therapy should be withheld at the onset of any Grade 3 non-hematological or Grade 4 hematological toxicities excluding neuropathy as discussed below (see Warnings and Precautions). Once the symptoms of the toxicity have resolved, CYTOMIB therapy may be reinitiated at a 25% reduced dose (1.3 mg/m^2/dose reduced to 1 mg/m^2/dose; 1 mg/m^2/dose reduced to 0.7 mg/m^2/dose).
For dose modifications guidelines for peripheral neuropathy, see Dose modifications for peripheral neuropathy below.

Dose Modifications for Peripheral Neuropathy.
Starting CYTOMIB subcutaneously may be considered for patients with pre-existing or at high risk of peripheral neuropathy. Patients with pre-existing severe neuropathy should be treated with CYTOMIB only after careful risk-benefit assessment.
Patients experiencing new or worsening peripheral neuropathy during CYTOMIB therapy may require a decrease in the dose and/or a less dose-intense schedule.
For dose or schedule modification guidelines for patients who experience CYTOMIB-related neuropathic pain and/or peripheral neuropathy see Table 6.

Table 6. Recommended Dose Modifications for CYTOMIB-Related Neuropathic Pain and/or Peripheral Sensory or Motor Neuropathy.

Severity of Peripheral Neuropathy Signs and Symptoms*	Modifications of dose and regime
Grade 1 (asymptomatic; loss of deep tendon reflexes or paresthesias) without pain or loss of function.	No action.
Grade 1 with pain or Grade 2 [moderate symptoms; limiting instrumental Activities of Daily Living (ADL)**]	Reduce CYTOMIB to 1 mg/m^2
Grade 2 with pain or Grade 3 (severe symptoms; limiting self care ADL***)	Withhold CYTOMIB therapy until toxicity resolves. When toxicity resolves reinitiate with a reduced dose of CYTOMIB at 0.7 mg/m^2 once per week.
Grade 4 (life- threatening consequences; urgent intervention indicated)	Discontinue CYTOMIB

*Grading based on NCI Common Toxicity Criteria CTCAE v4.0.
**Instrumental ADL: refers to preparing meals, shopping for groceries or clothes, using telephone, managing money etc.
***Self care ADL: refers to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and not bedridden

Special Populations
Patients with Renal Impairment
The pharmacokinetics of CYTOMIB are not influenced by the degree of renal impairment. Therefore, dosing adjustments of CYTOMIB are not necessary for patients with renal insufficiency. Since dialysis may reduce CYTOMIB concentrations, the drug should be administered after the dialysis procedure (see Pharmacokinetic Properties).
Patients with Hepatic Impairment

Patients with mild hepatic impairment do not require a starting dose adjustment and should be treated per the recommended CYTOMIB dose. For patients with moderate or severe hepatic impairment, see Table 7 below, (also, see Pharmacokinetic Properties):

Table 7. Recommended Starting Dose Modification for CYTOMIB in Patients with Hepatic Impairment.

	Bilirubin Level	SGOT (AST) Levels	Modification of Starting Dose in Multiple Myeloma and Relapsed Mantle Cell Lymphoma (1.3 mg/ m^2 twice weekly)
Mild	≤ 1.0x ULN	> ULN	None
	> 1.0x - 1.5x ULN	Any	None
Moderate	> 1.5x - 3x ULN	Any	Reduce CYTOMIB to 0.7 mg/m^2 in the 1st cycle. Consider dose escalation to 1 mg/m^2 or further dose reduction to 0.5 mg/m^2 in subsequent cycles based on patient tolerability.
Severe	> 3x ULN	Any	
SGOT = serum glutamic oxaloacetic transaminase; AST = aspartate aminotransferase; ULN = upper limit of the normal range.			

Administration: CYTOMIB is administered intravenously or subcutaneously. When administered intravenously, CYTOMIB is administered as a 3-5 second bolus intravenous injection through a peripheral or central intravenous catheter followed by a flush with 0.9% sodium chloride solution for injection. For subcutaneous administration, the reconstituted solution is injected into the thighs (right or left) or abdomen (right or left). Injection sites should be rotated for successive injections.
If local injection site reactions occur following CYTOMIB injection subcutaneously, a less concentrated CYTOMIB solution (1 mg/ml instead of 2.5 mg/ml) may be administered subcutaneously, or changed to IV injection.

CONTRAINDICATIONS
CYTOMIB is contraindicated in patients with hypersensitivity to bortezomib, boron or mannitol.

WARNING AND PRECAUTIONS
CYTOMIB should be administered under the supervision of a physician experienced in the use of antineoplastic therapy.
There have been fatal cases of inadvertent intrathecal administration of Bortezomib. CYTOMIB is for IV and subcutaneous use only. DO NOT ADMINISTER CYTOMIB INTRATHECALLY.
Peripheral Neuropathy
CYTOMIB treatment causes a peripheral neuropathy (PN) that is predominantly sensory. However, cases of severe motor neuropathy with or without sensory peripheral neuropathy have been reported.
Patients with pre-existing symptoms (numbness, pain or a burning feeling in the feet or hands) and/or signs of peripheral neuropathy may experience worsening peripheral neuropathy (including ≥Grade 3) during treatment with CYTOMIB. Patients should be monitored for symptoms of neuropathy eg, a burning sensation, hyperesthesia, hypoesthesia, paresthesia, discomfort, neuropathic pain or weakness.
Patients experiencing new or worsening peripheral neuropathy may require change in the dose and schedule of CYTOMIB. Patients experiencing new or worsening peripheral neuropathy may require a change in dose schedule or route of administration to SC (see Recommended dose and instruction for use).
The long-term outcome of peripheral neuropathy has not been studied in mantle cell lymphoma.
Hypotension
The incidence of hypotension (postural, orthostatic, and hypotension not otherwise specified) was 11-12%. These events are observed throughout therapy. Caution should be used when treating patients with a history of syncope, patients receiving medications known to be associated with hypotension and patients who are dehydrated. Management of orthostatic/postural hypotension may include adjustment of antihypertensive medications, hydration or administration of mineralocorticoids and/or sympathomimetics (See Side effects).
Cardiac Disorders
Acute development or exacerbation of congestive heart failure, and/or new onset of decreased left ventricular ejection fraction has been reported, including reports in patients with few or no risk factors for decreased left ventricular ejection fraction. Patients with risk factors for, or existing heart disease should be closely monitored. The incidence of heart failure events (acute pulmonary edema, cardiac failure, congestive cardiac failure, cardiogenic shock, pulmonary edema) was similar in the CYTOMIB and dexamethasone-group.
Hepatic Events
Rare cases of acute liver failure have been reported in patients receiving multiple concomitant medications and with serious underlying medical conditions. Other reported hepatic events include increases in liver enzymes, hyperbilirubinemia and hepatitis. Such changes may be reversible upon discontinuation of CYTOMIB.
Pulmonary Disorders
There have been rare reports of acute diffuse infiltrative pulmonary disease of unknown etiology e.g., pneumonitis, interstitial pneumonia, lung infiltration and acute respiratory distress syndrome (ARDS) in patients receiving CYTOMIB. Some of these events have been fatal. In the event of new or worsening pulmonary symptoms, a prompt diagnostic evaluation should be performed and patients treated appropriately.
Laboratory Tests
Complete blood counts (CBC) should be frequently monitored throughout treatment with CYTOMIB.
Thrombocytopenia/Neutropenia
CYTOMIB is associated with thrombocytopenia and neutropenia (see Side Effects). Platelets were lowest at Day 11 of each cycle of CYTOMIB treatment and typically recovered to baseline

by the next cycle. The cyclical pattern of platelet count decrease and recovery remain consistent in multiple myeloma and mantle cell lymphoma, with no evidence of cumulative thrombocytopenia or neutropenia in any of the regimens studied.
Platelet counts should be monitored prior to each dose of CYTOMIB. CYTOMIB therapy should be held when the platelet count is <25,000/uL (see **Recommended dose, Instruction for Use and Side Effects**). There have been reports of gastrointestinal and intracerebral hemorrhage in association with CYTOMIB. Transfusion and supportive care may be considered.
Gastrointestinal Adverse Events
CYTOMIB treatment can cause nausea, diarrhoea, constipation, and vomiting (see Side Effects) sometimes requiring use of antiemetics and antidiarrheal medications. Fluid and electrolyte replacement should be administered to prevent dehydration. Since patients receiving CYTOMIB therapy may experience vomiting and/or diarrhoea, patients should be advised regarding appropriate measures to avoid dehydration. Patients should be instructed to seek medical advice if they experience symptoms of dizziness, light headedness or fainting spells.
Tumour Lysis Syndrome
Because CYTOMIB is a cytotoxic agent and can rapidly kill malignant cells, the complications of tumour lysis syndrome may occur. The patients at risk of tumour lysis syndrome are those with high-tumour burden prior to treatment. These patients should be monitored closely and appropriate precautions taken.
Patients with Hepatic Impairment
Bortezomib is metabolized by liver enzymes. Bortezomib exposure is increased in patients with moderate or severe hepatic impairment; these patients should be treated with CYTOMIB at reduced starting doses and closely monitored for toxicities (See **Recommended Dose and Pharmacokinetic properties**).
Posterior Reversible Encephalopathy Syndrome (PRES)
There have been reports of PRES in patients receiving CYTOMIB. PRES is a rare, reversible, neurological disorder which can present with seizure, hypertension, headache, lethargy, confusion, blindness, and other visual and neurological disturbances. Brain imaging, preferably magnetic resonance imaging (MRI), is used to confirm the diagnosis. In patients developing PRES, discontinue CYTOMIB. The safety of reinitiating CYTOMIB therapy in patients previously experiencing PRES is not known.

INTERACTIONS WITH OTHER MEDICAMENTS
Bortezomib is a weak inhibitor of cytochrome P-450 (CYP) isozymes 1A2, 2C9, 2C19, 2D6 and 3A4. Based on the limited contributionof CYP2D6 to the metabolism of bortezomib, the CYP2D6 poor metabolizer phenotype is not expected to affect the overall disposition of bortezomib.
A drug-drug interaction study assessing the effect of ketoconazole, a potent CYP3A4 inhibitor, showed a bortezomib AUC mean increase of 35%. Therefore, patients should be closely monitored when given bortezomib in combination with potent CYP3A4 inhibitors (eg, ketoconazole, ritonavir).
In a drug-drug interaction study assessing the effect of omeprazole, a potent inhibitor of CYP2C19, there was no significant effect on the pharmacokinetics of bortezomib.
A drug-drug interaction study assessing the effect of melphalan-prednisone on CYTOMIB showed an increase in mean bortezomib AUC. This is not considered clinically relevant.
Hypoglycemia and hyperglycemia were reported in diabetic patients receiving oral hypoglycemics. Patients on oral antidiabetic agents receiving CYTOMIB treatment may require close monitoring of their blood glucose levels and adjustment of the dose of their antidiabetic medication.
Patients should be cautioned about the use of concomitant medications that may be associated with peripheral neuropathy (eg, amiodarone, antivirals, isoniazid, nitrofurantoin or statins), or with a decrease in blood pressure.
Drug Laboratory Interactions: None known.

PREGNANCY & LACTATION
Women of childbearing potential should avoid becoming pregnant while being treated with CYTOMIB. Bortezomib was not teratogenic in nonclinical developmental toxicity studies.
No placental transfer studies have been conducted with bortezomib. There are no adequate and well-controlled studies in pregnant women. If CYTOMIB is used during pregnancy, or if the patient becomes pregnant while receiving this drug, the patient should be apprised of the potential hazard to the foetus.
Patients should be advised to use effective contraceptive measures to prevent pregnancy and to avoid breastfeeding during treatment with CYTOMIB.
Breast-feeding
It is not known whether bortezomib is excreted in human milk. Because many drugs are excreted in human milk and because of the potential for serious adverse reactions in nursing infants from CYTOMIB, women should be advised against breastfeeding while being treated with CYTOMIB.

SIDE EFFECTS
Summary of Patients with Previously Untreated Multiple Myeloma: Table 5 describes safety data from patients with previously untreated multiple myeloma who received bortezomib (1.3 mg/m^2) in combination with melphalan (9 mg/m^2) and prednisone (60 mg/m^2).

Table 5: Treatment Emergent Drug-Related Adverse Events Reported in Patients Treated with Bortezomib in Combination with Melphalan and Prednisone
MedDRA System Organ Class Preferred Term
Blood and Lymphatic system disorders
• Thrombocytopenia
• Neutropenia • Anaemia • Leukopenia • Lymphopenia
Gastrointestinal disorders
• Nausea
• Diarrhoea • Vomiting • Constipation • Upper abdominal pain
Nervous system disorders
• Peripheral neuropathy • Neuralgia • Paraesthesia • Guillain-Barré syndrome (Frequency 'rare') • Demyelinating polyneuropathy (Frequency 'rare')
General disorders and administration site conditions
• Fatigue • Asthenia • Pyrexia
Infections and infestations
• Herpes zoster
Metabolism and nutritional disorders
• Anorexia
Skin and subcutaneous tissue disorders
• Rash
Psychiatric disorders
• Insomnia

Herpes-Zoster Virus Reactivation
Physician should consider using antiviral prophylaxis in patients being treated with CYTOMIB. In patients with previously untreated multiple myeloma, the overall incidence of herpes zoster reactivation was more common in patients treated with CYTOMIB in combination with Melphalan and Prednisone compared with combination of Melphalan and Prednisone. Antiviral prophylaxis was administered to patients in the CYTOMIB in combination with Melphalan and Prednisone arm.
Patients with Mantle Cell Lymphoma
Safety data for patients with mantle cell lymphoma were evaluated in patients treated with CYTOMIB at the recommended dose of 1.3 mg/m^2. The safety profile of CYTOMIB in these patients was similar to that observed in patients with multiple myeloma. Notable differences between the patient populations were that thrombocytopenia, neutropenia, anemia, nausea, vomiting and pyrexia were reported more often in the patients with multiple myeloma than in those with mantle cell lymphoma; whereas peripheral neuropathy, rash and pruritus were higher among patients with mantle cell lymphoma compared to patients with multiple myeloma.
Post-Marketing Experience
Clinically significant adverse drug reactions are listed as follows if they have not been reported previously.
The frequencies provided reflect reporting rates of adverse drug reactions from the worldwide post-marketing experience with CYTOMIB and precise estimates of incidence cannot be made.
Blood and Lymphatic System Disorders: Rare: Disseminated intravascular coagulation.
Cardiac Disorders: Rare: Complete atrioventricular block, cardiac tamponade.
Ear and Labyrinth Disorders: Rare: Bilateral deafness.
Eye Disorders: Rare: Ophthalmic herpes.
Gastrointestinal Disorders: Rare: Ischemic colitis, acute pancreatitis.
Infections and Infestations: Rare: Herpes meningoencephalitis, septic shock.
Immune System Disorders: Rare: Angioedema.
Nervous System Disorders: Rare: Encephalopathy, autonomic neuropathy, reversible posterior leukoencephalopathy syndrome.
Respiratory, Thoracic and Mediastinal Disorders: Rare: Acute diffuse infiltrative pulmonary disease and pulmonary hypertension.
Skin and Subcutaneous Tissue Disorders: Rare: Acute febrile neutrophilic dermatosis (Sweet's syndrome). Very Rare: Stevens-Johnson Syndrome and toxic epidermal necrolysis.

SYMPTOMS AND TREATMENT OF OVERDOSE
Cardiovascular safety pharmacology studies show that IV doses approximately 2-3 times the recommended clinical dose on a mg/m^2 basis are associated with increases in heart rate, decreases in contractility, hypotension and death. The decreased cardiac contractility and hypotension responded to acute intervention with positive inotropic or pressor agents. A slight increase in the corrected QT interval was observed at a lethal dose.
Overdosage more than twice the recommended dose in patients has been associated with the acute onset of symptomatic hypotension and thrombocytopenia with fatal outcomes.
There is no known specific antidote for CYTOMIB overdosage. In the event of an overdosage, the patient's vital signs should be monitored, and appropriate supportive care given to maintain blood pressure (eg, fluids, pressors and/or inotropic agents) and body temperature.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE
CYTOMIB may cause tiredness, dizziness, fainting, or blurred vision. Patients should be advised not to drive or operate machinery if they experience these symptoms.

INSTRUCTIONS FOR USE
CYTOMIB should be administered under the supervision of a physician experienced in the use of antineoplastic therapy.
Administration Precautions:
CYTOMIB is an antineoplastic. Caution should be used during handling and preparation. Proper aseptic technique should be used. Use of gloves and other protective clothing to prevent skin contact is recommended. Local skin irritation was reported in patients, but extravasation of CYTOMIB was not associated with tissue damage.
When administered subcutaneously, alternate sites for each injection (thigh or abdomen). New injections should be given at least one inch from an old site and never into areas where the site is tender, bruised, red, or hard
There have been fatal cases of inadvertent intrathecal administration of CYTOMIB. CYTOMIB is for IV and subcutaneous use only. **DO NOT ADMINISTER CYTOMIB INTRATHECALLY.**

Reconstitution/Preparation for Intravenous and Subcutaneous Administration
The contents of each vial should be reconstituted only with normal (0.9%) saline according to the following instructions based on route of administration:

	IV	SC
	3.5 mg bortezomib	3.5 mg bortezomib
Volume of diluent (0.9% Sodium Chloride) added to reconstitute one vial.	3.5 mL	1.4 mL
Final Concentration after reconstitution (mg/mL).	1.0 mg/mL	2.5 mg/mL

The reconstituted product should be a clear and colorless solution. Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration whenever solution and container permit. If any discoloration or particulate matter is observed, the reconstituted product should not be used.
Disposal:
CYTOMIB is for single use only. Any unused medicinal product or waste material should be disposed of in accordance with local requirements
Incompatibilities:
CYTOMIB must not be mixed with other medicinal products except those mentioned in Instruction for Use.

STORAGE CONDITION
Before reconstitution:
Store below 30°C. Protect from light. Keep the vial in the outer carton in order to protect from light.
After reconstitution:
The reconstituted solution should be used immediately after preparation. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user. However, the chemical and physical in-use stability of the reconstituted solution has been demonstrated for 8 hours at 25°C stored in the original vial and/or a syringe. The total storage time for the reconstituted medicinal product should not exceed 8 hours prior to administration.

SHELF LIFE
Unopened vial:
36 Months from date of manufacturing.
Reconstituted solution:
The reconstituted solution should be used immediately after preparation. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user. However, the chemical and physical in-use stability of the reconstituted solution has been demonstrated for 8 hours at 25°C stored in the original vial and/or a syringe. The total storage time for the reconstituted medicinal product should not exceed 8 hours prior to administration.

PRESENTATION / PACKING
CYTOMIB (Bortezomib 3.5 mg Powder for Solution for Injection), sterile white to off-white lyophilized mass.in single-dose vial is packaged in one 10 ml Type I clear tubular glass vial, in unit carton along with package insert (without diluent).

Manufactured by:
VENUS REMEDIES LIMITED
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